Cluster Analysis

Discovering Hidden Groups and Structures in Data

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Introduction to Cluster Analysis

Clustering

- An unsupervised learning technique.
- Goal: To group a set of data objects into clusters.

The Golden Rule of Clustering:

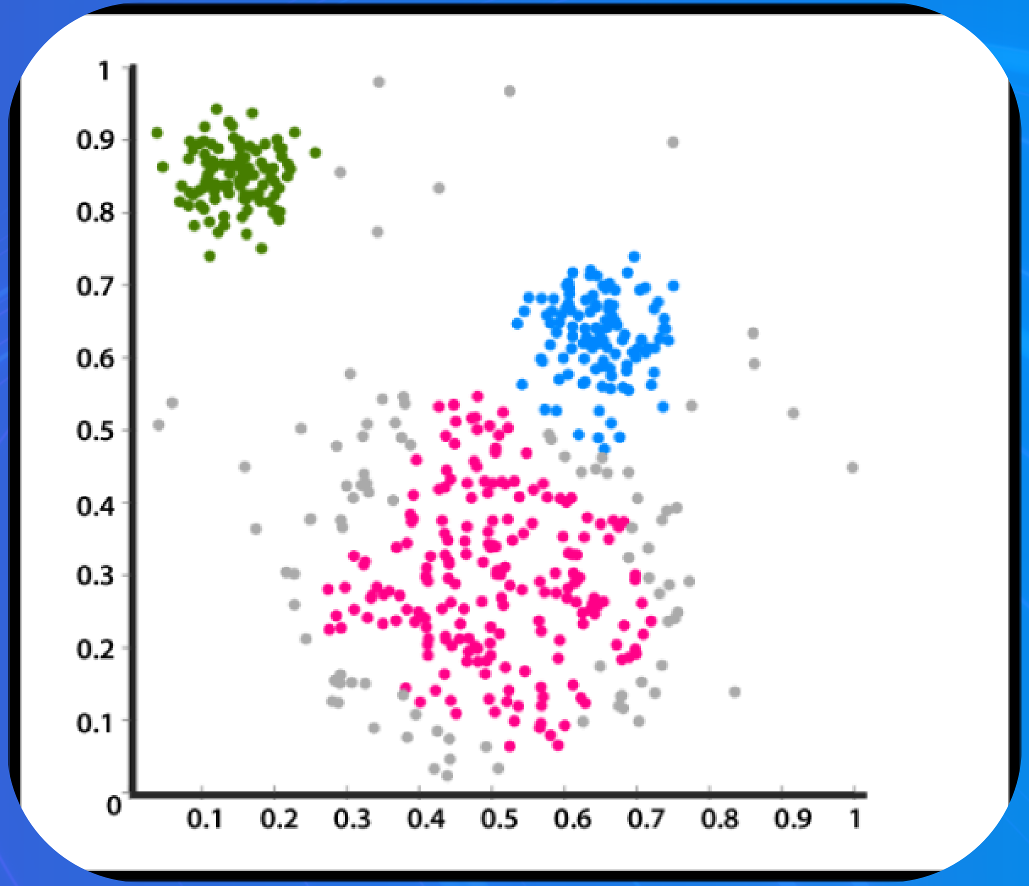
- Objects within a cluster should be highly similar.
- Objects in different clusters should be highly dissimilar.

Why is it "Unsupervised"?

- Because the algorithm is not given any predefined labels. It finds the structure entirely on its own.

Common Applications:

- Marketing: Segmenting customers into different groups for targeted campaigns.
- Biology: Grouping genes with similar expression patterns.
- Image Processing: Segmenting an image into different regions.



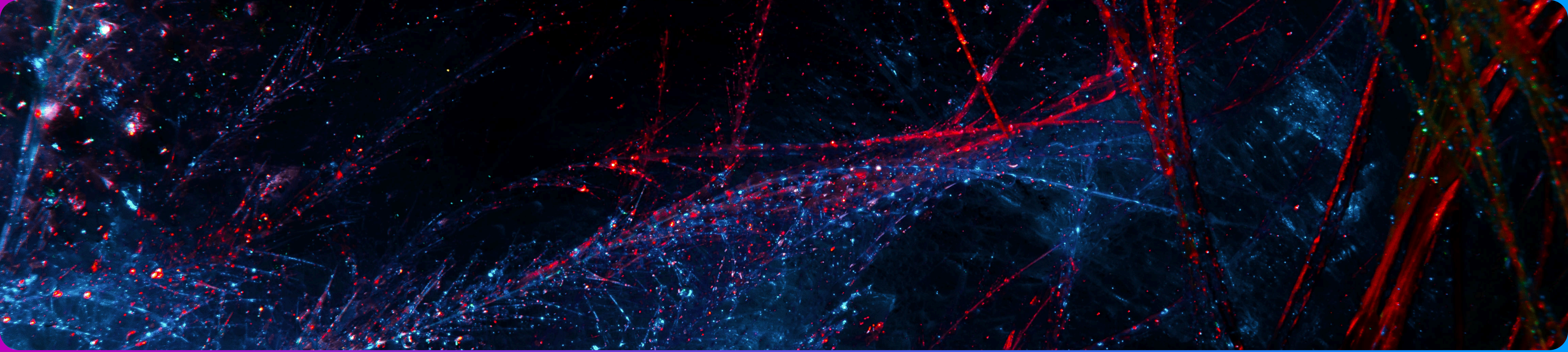
Partitioning Methods

Dividing Data into k Distinct Groups

- Concept: These methods break the dataset down into a pre-specified number of non-overlapping clusters (k).
- The algorithm iteratively rearranges data points to find the best possible partitions.

K-Means Algorithm:

- The most popular partitioning method.
- Represents each cluster by its centroid (the average of all points in the cluster).
- Process:
  - Randomly select k initial centroids.
  - Assign each data point to its closest centroid.
  - Recalculate the centroids based on the new cluster members.
  - Repeat steps 2-3 until the clusters stabilize.
- Challenge: You must choose the value of k in advance.



Hierarchical Methods

- This method creates a nested hierarchy of clusters, which can be visualized as a dendrogram .
- Advantage: You do not need to specify the number of clusters beforehand.

Two Main Strategies:

- Agglomerative (Bottom-up):
  - Starts with each data point as an individual cluster.
  - At each step, it merges the two closest clusters.
  - Continues until all points are in a single cluster.
- Divisive (Top-down):
  - Starts with all data points in one giant cluster.
  - At each step, it splits a cluster into two.
  - Continues until each point is its own cluster.

Advanced Density and Grid-Based Methods

DBSCAN (Density-Based Spatial Clustering)

- A cluster is a dense region of points, separated from other clusters by low-density regions.
- It defines clusters based on the "density reachability" of points.
- Strengths:
  - Can discover clusters of non-spherical shapes.
  - Is robust to outliers, which it identifies as "noise."

Grid-Based Methods (e.g., STING, CLIQUE)

- The data space is divided into a grid structure. All clustering operations are performed on this grid.
- CLIQUE: Finds clusters in subspaces, making it effective for high-dimensional data.
- Advantage: Very fast, as its speed depends on the grid size, not the number of data points.

Probabilistic and Large-Scale Clustering

Probabilistic Model-Based Clustering

- The data is a mixture of several underlying probability distributions (e.g., Gaussian distributions).
- The algorithm tries to find the distribution parameters that best fit the data.
- Expectation-Maximization (EM): A common algorithm used to find these parameters. It provides a "soft" clustering, where each point has a probability of belonging to each cluster.

BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies)

- A highly scalable algorithm.
- How it works:
- It first scans the database to create a compact, in-memory summary called a CF-Tree.
- It then applies a standard clustering algorithm to the leaf nodes of this tree.
- Advantage: Fast and memory-efficient.

Evaluation of Clustering Techniques

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Internal Evaluation (Unsupervised)

- Assesses the quality of the clustering structure without using external information.
- Measures:
  - Cohesion: How tightly packed is each cluster? (e.g., Within-cluster sum of squares).
  - Separation: How well-separated are the different clusters?
  - Silhouette Coefficient: A single score (from -1 to 1) that combines both cohesion and separation. A higher score is better.

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External Evaluation (Supervised)

- Used when we have ground truth labels to compare against.
- Purity: Measures the extent to which a cluster contains a single class.
- Rand Index: Measures the similarity between two data clusterings.

Thank You